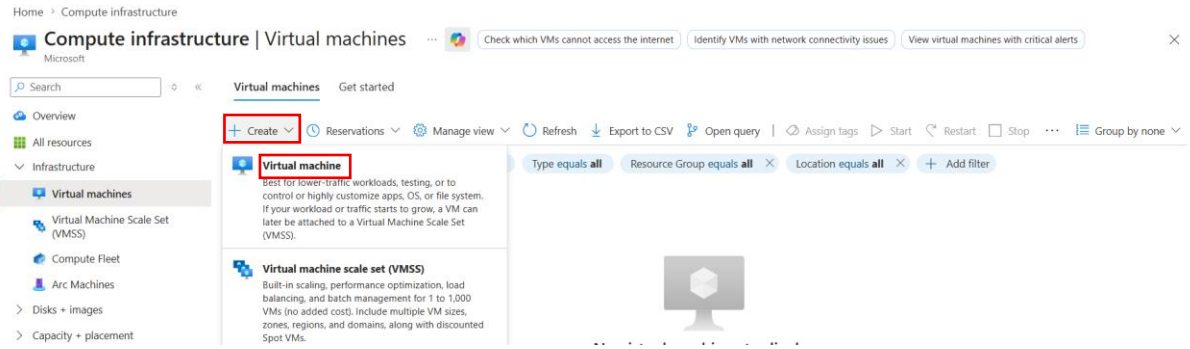
Docker with Azure

To Begin with the Lab

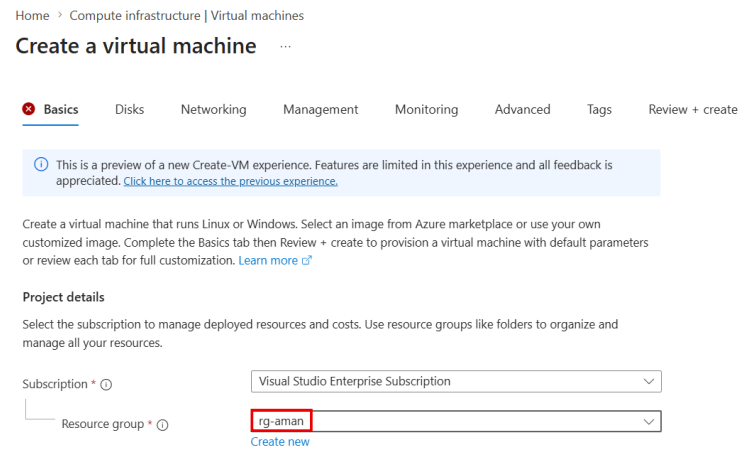
Summary of the Lab

This lab demonstrated how to deploy and manage Docker applications on Microsoft Azure using multiple Azure services. An Ubuntu Linux Virtual Machine was created, Docker and Docker Compose were installed, and MySQL and a PHP application were deployed using Docker Compose. The application database was configured with course data, while course images were migrated from local storage to Azure Blob Storage and accessed through Blob URLs. The Docker images were then pushed to Azure Container Registry (ACR) for centralized image management. Finally, the MySQL container was deployed using Azure Container Instances (ACI), demonstrating how to run containerized applications without managing virtual machines.

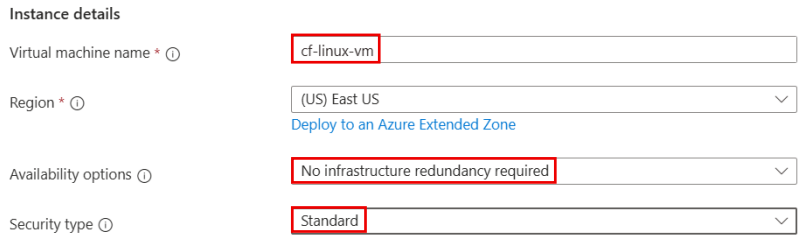
- Open the **Azure Portal** and navigate to **Virtual Machines**, then click **Create > Azure Virtual Machine**.



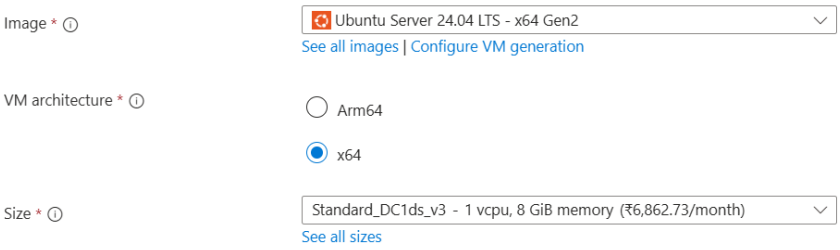
- Select your **Azure Subscription**, create or choose a **Resource Group**, and proceed to the virtual machine configuration page.
- Also, you can create your own.



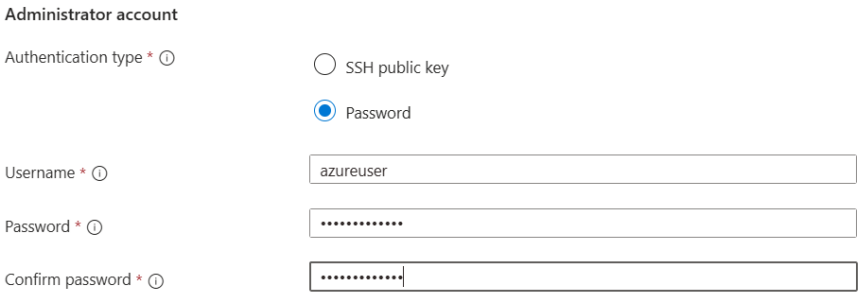
- Enter a **Virtual Machine Name**, choose the desired **Region**, and set **Availability Options** to **No infrastructure redundancy required**.



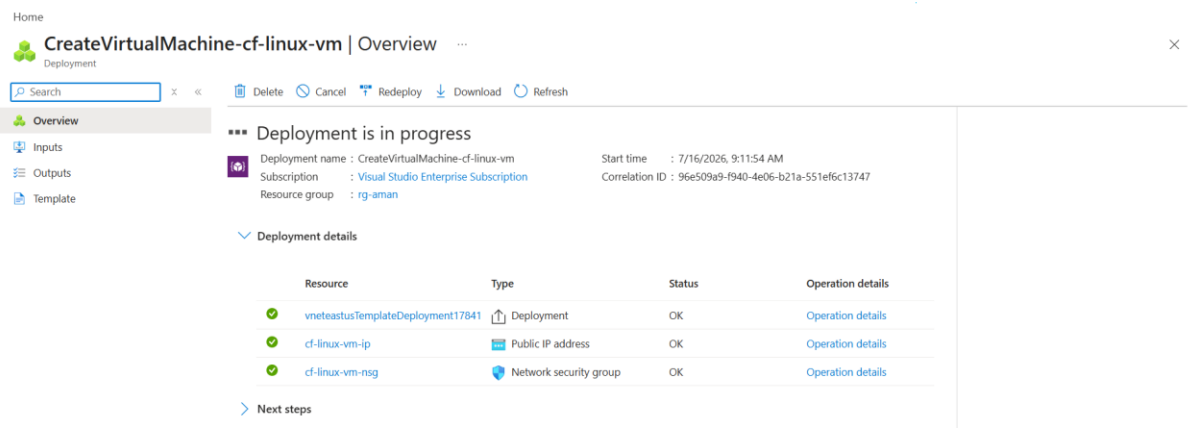
- Under **Image**, select **Ubuntu Server 22.04 LTS (Gen 2)**. If it is not visible, click **See all images**, search for **Ubuntu**, and select the required image.
- Keep the default VM size (**Standard_B1s**) or choose another size if required, then continue.



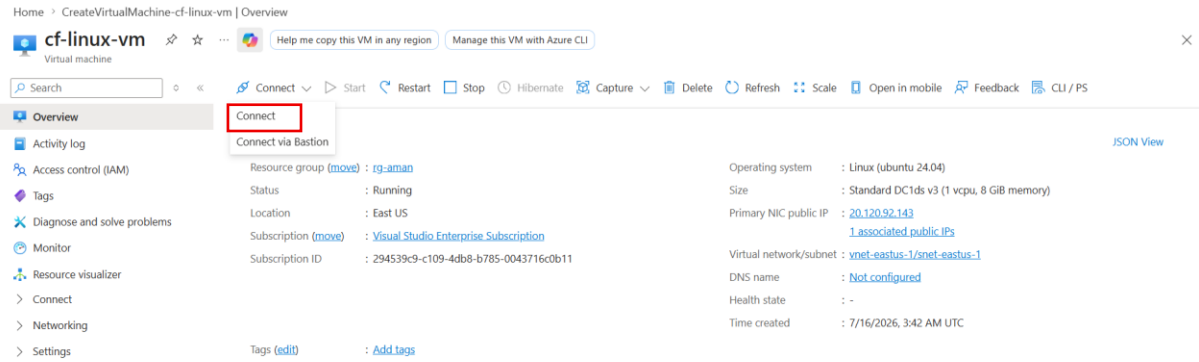
- Select **Password** as the authentication method and provide a valid **Username** and **Password** that meet Azure's password requirements.



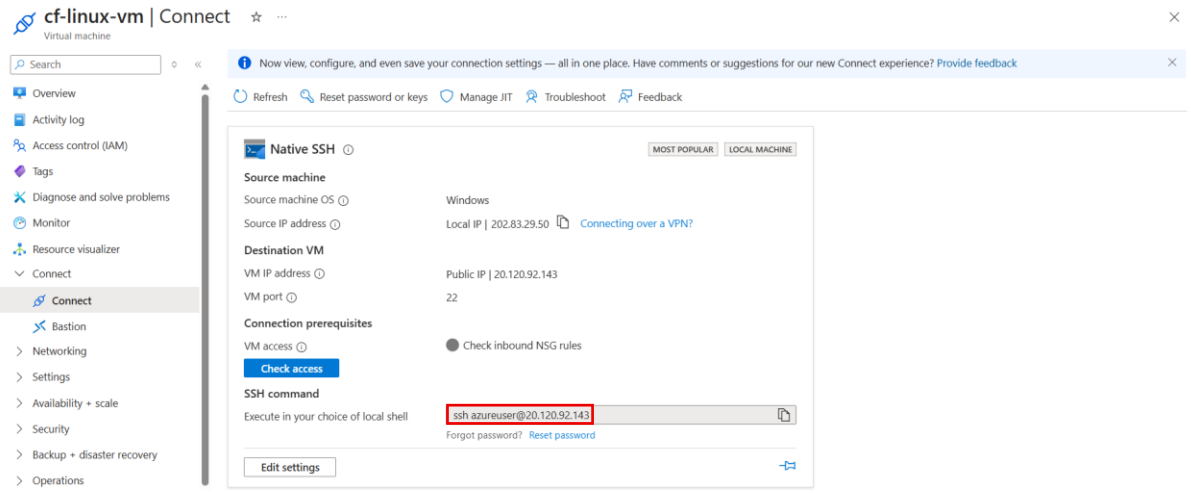
- Click **Next: Disks** and keep the default OS disk configuration.
- Click **Next: Networking** and keep the default settings, allowing Azure to create the virtual network, public IP address, network interface, and network security group automatically.
- Continue through the **Management**, **Monitoring**, **Advanced**, and **Tags** tabs without making any changes.
- On the **Review + Create** page, verify the configuration and estimated hourly cost, then click **Create**.
- Wait for the deployment to complete, then click **Go to Resource** to open the newly created virtual machine.
- Open **All Resources** to verify that the deployment has created the **Virtual Machine**, **Public IP Address**, **Virtual Network**, **Network Interface**, **Network Security Group**, **OS Disk**, and other supporting resources successfully.



- After the resources are created successfully
- Go to your virtual machine and click connect



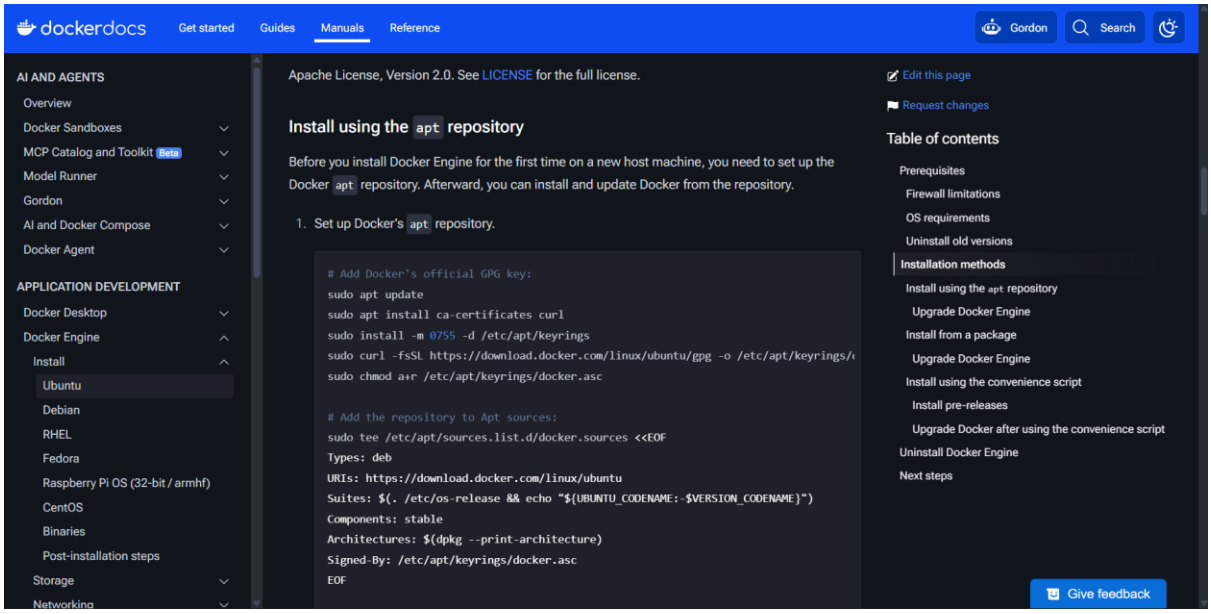
- Copy the SSH command



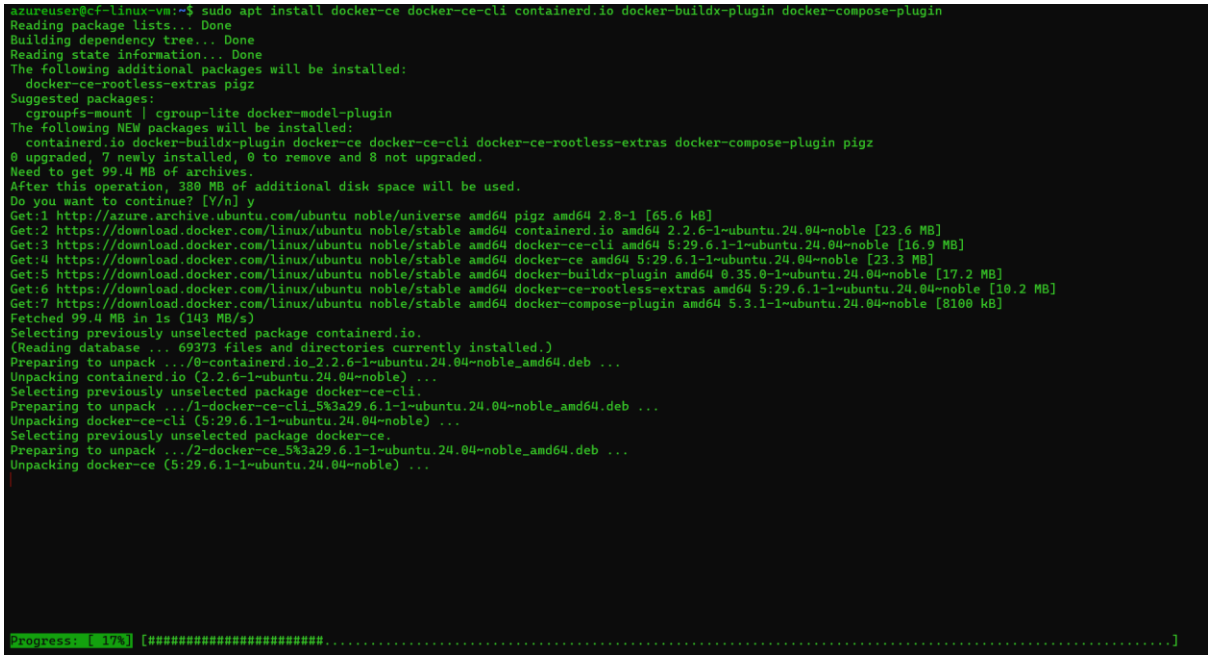
- Open the command prompt or terminal on the local system and login through the command

```
C:\Users\amanr>ssh azureuser@20.120.92.143
azureuser@20.120.92.143's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1020-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Jul 16 03:48:56 UTC 2026

  System load:  0.0               Processes:             113
  Usage of /:   5.7% of 28.02GB   Users logged in:       0
  Memory usage: 3%                IPv4 address for eth0: 172.16.0.4
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Thu Jul 16 03:44:28 2026 from 202.83.29.50
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@cf-linux-vm:~$
```

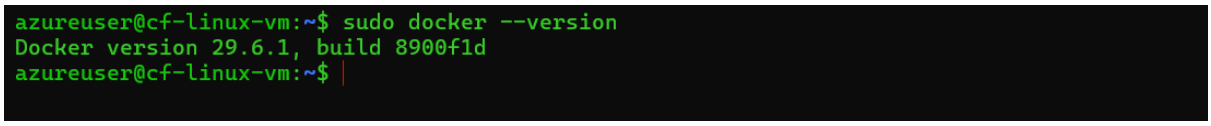
- Go to docker official webpage get the step to install the docker



- After running all the command docker is installing



- Let's check the docker version



Running a MySQL Docker Container on the Azure Linux VM

- Verify that no Docker images are available on the Linux VM:

```
sudo docker images
```

- Run a MySQL container by pulling the image from Docker Hub and publishing port **3306**:

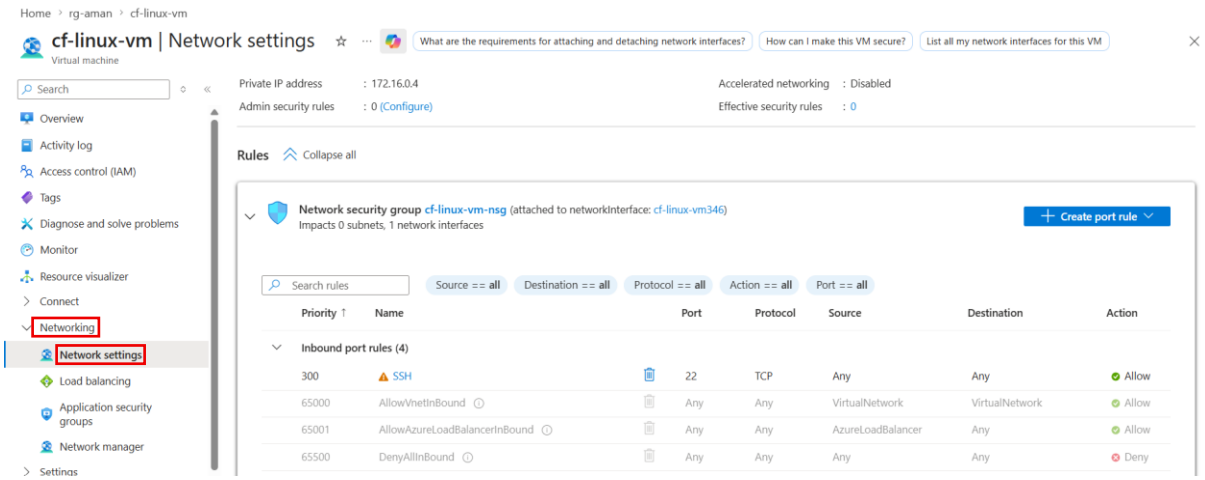
```
sudo docker run -d -p 3306:3306 --name mysql-server -e MYSQL_ROOT_PASSWORD=1234 mysql
```

- Verify that the MySQL container is running:

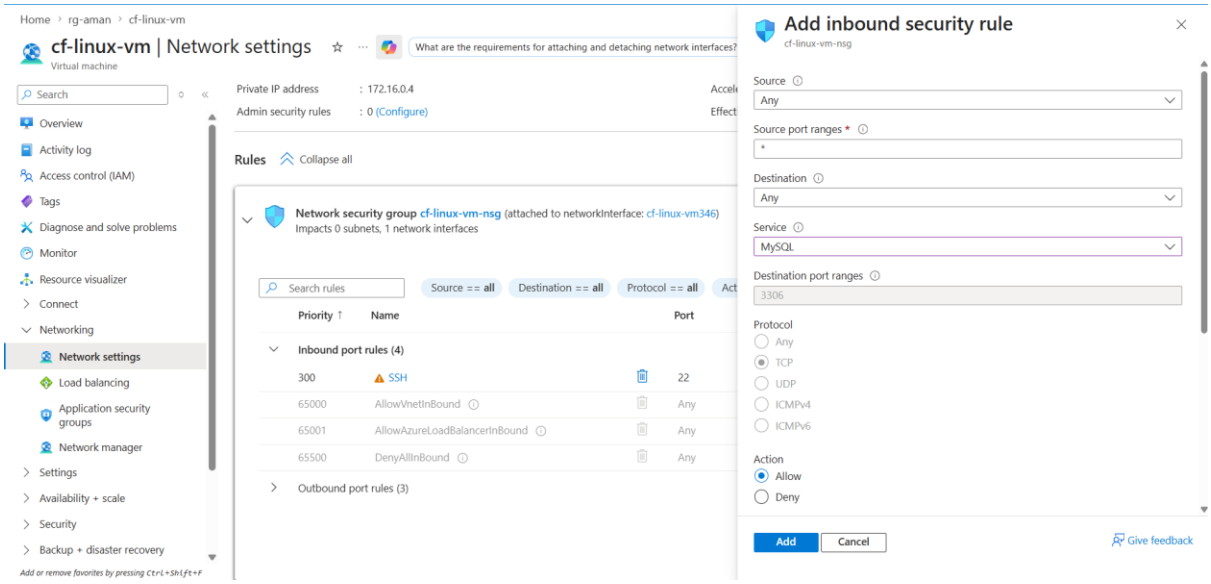
```
sudo docker ps
```

```
azureuser@cf-linux-vm:~$ sudo docker images
IMAGE   ID             DISK USAGE   CONTENT SIZE   EXTRA
azureuser@cf-linux-vm:~$ sudo docker run -d -p 3306:3306 --name mysql-server -e MYSQL_ROOT_PASSWORD=1234 mysql
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
446597dc68d2: Pull complete
ecf47111addb: Pull complete
d4c51daaa784: Pull complete
ad18077cf381: Pull complete
9f5b662d2d4d: Pull complete
890dd41ed99a: Pull complete
1257e53bba2a: Pull complete
6b6fad81c717: Pull complete
aeff22d41055: Pull complete
04eb080c2a10: Pull complete
62f7e05598ba: Download complete
54e5e996b461: Download complete
Digest: sha256:ae269281abffe401d65f04cb54d45a869a495b8174b9c0a815e502fed7fa0370
Status: Downloaded newer image for mysql:latest
1ea6c3fb8a78c3c41082c988f632795e3161dce18c40f017d05ba1046b0dffa2
azureuser@cf-linux-vm:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED              STATUS              PORTS                                                  NAMES
1ea6c3fb8a78   mysql     "docker-entrypoint.s…"   About a minute ago   Up About a minute   0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp, 33060/tcp   mysql-server
azureuser@cf-linux-vm:~$
```

- In the **Azure Portal**, open the **Virtual Machine**, navigate to **Networking**, and click **Add inbound port rule**.

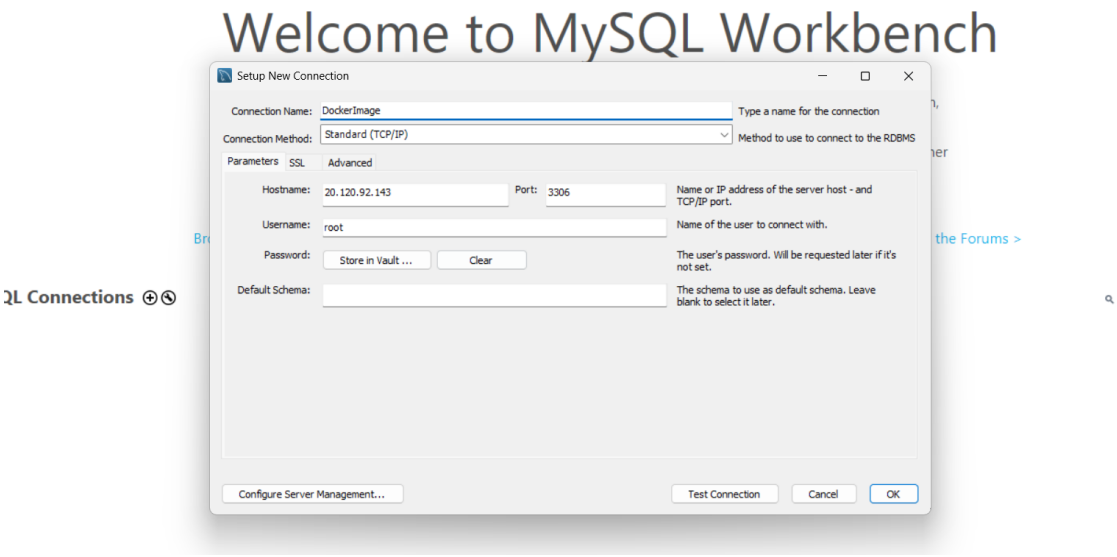


- Select the **MySQL (TCP 3306)** service (or manually allow **TCP port 3306**) and create the inbound rule.

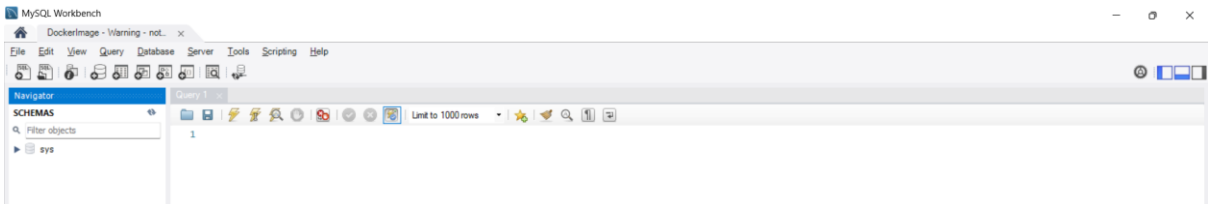


- Copy the **Public IP Address** of the virtual machine from the **Overview** page.

- Open **MySQL Workbench** (or any MySQL client), create/edit a connection, and use the VM's **Public IP Address** as the hostname.
- Enter the **root** username and the password specified while creating the container, then connect to verify the MySQL server is accessible.



- Confirm that the MySQL database is running successfully on the Docker container hosted inside the Azure Linux Virtual Machine.



Running Docker Compose on the Azure Linux VM

- Navigate to the application directory containing the Dockerfile, docker-compose.yaml, PHP application, and Bootstrap files.
- Verify that no containers are currently running:

```
sudo docker ps -a
```

- Create a new Docker Compose file on the VM:

```
cat > docker-compose.yaml
```

- Create or copy the docker-compose.yaml file to the Linux VM and verify its contents.
- Verify that the file has been created:

```
ls
```

View the contents of the Docker Compose file:

cat docker-compose.yaml

- code

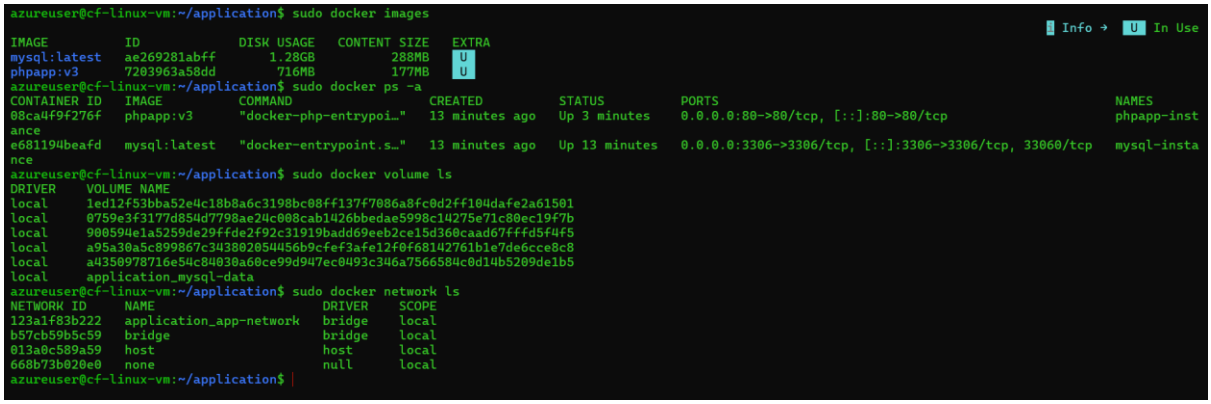
```
services:
  webapp:
    image: phpapp:v3
    build: .
    container_name: phpapp-instance
    depends_on:
      - db
    networks:
      - app-network
    ports:
      - "80:80"

  db:
    image: mysql:latest
    container_name: mysql-instance
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: sqlset1010
      MYSQL_DATABASE: appdb
      MYSQL_USER: appusr
      MYSQL_PASSWORD: appusr@1090
    ports:
      - "3306:3306"
    volumes:
      - mysql-data:/var/lib/mysql
    networks:
      - app-network
```

```
networks:
  app-network:
    driver: bridge
    ipam:
      config:
        - subnet: 172.28.0.0/16

volumes:
  mysql-data:
```

- Build and deploy the application using Docker Compose.

```
sudo docker compose up -d –build
```

- Verify that the Docker image, custom network, volume, and both containers are created successfully.



- Enter the MySQL container

```
sudo docker exec -it mysql-instance mysql -uroot -psqlset1010
```

- Before creating table, check databases

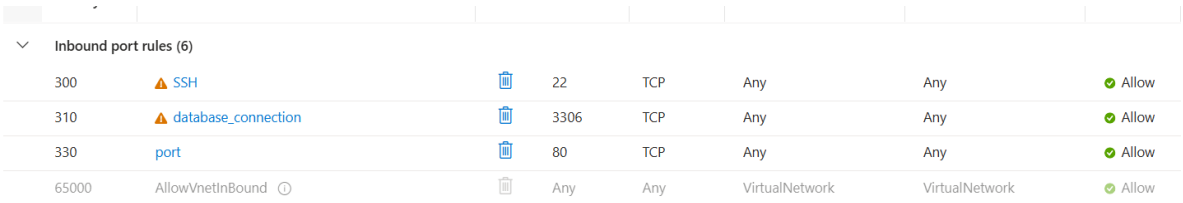
```
show databases;
```

- check for appdb
- Create the table, Inside MySQL, execute:

```
USE appdb;
```

```
CREATE TABLE Course (
    CourseID INT PRIMARY KEY,
    ExamImage VARCHAR(255),
    CourseName VARCHAR(100),
    rating DECIMAL(2,1)
);

INSERT INTO Course VALUES
(101,'https://via.placeholder.com/400x100','Docker Fundamentals',4.8),
(102,'https://via.placeholder.com/400x100','Docker Compose',4.7),
(103,'https://via.placeholder.com/400x100','Azure for Containers',4.9),
(104,'https://via.placeholder.com/400x100','Kubernetes Basics',4.6);

GRANT ALL PRIVILEGES ON appdb.* TO 'appusr'@'%';
FLUSH PRIVILEGES;
EXIT;
```

- In the Azure Portal, open the VM's Networking settings and add an HTTP (TCP Port 80) inbound rule.



- Copy the VM's Public IP Address and open it in a browser to verify that the PHP application is running successfully.

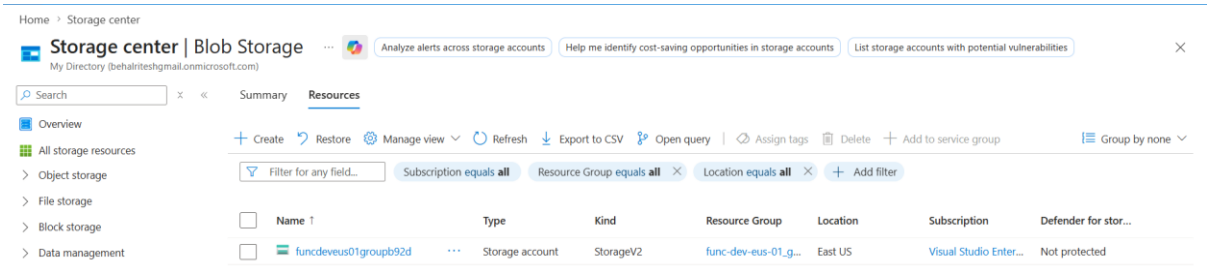
Courses

This is a list of Courses

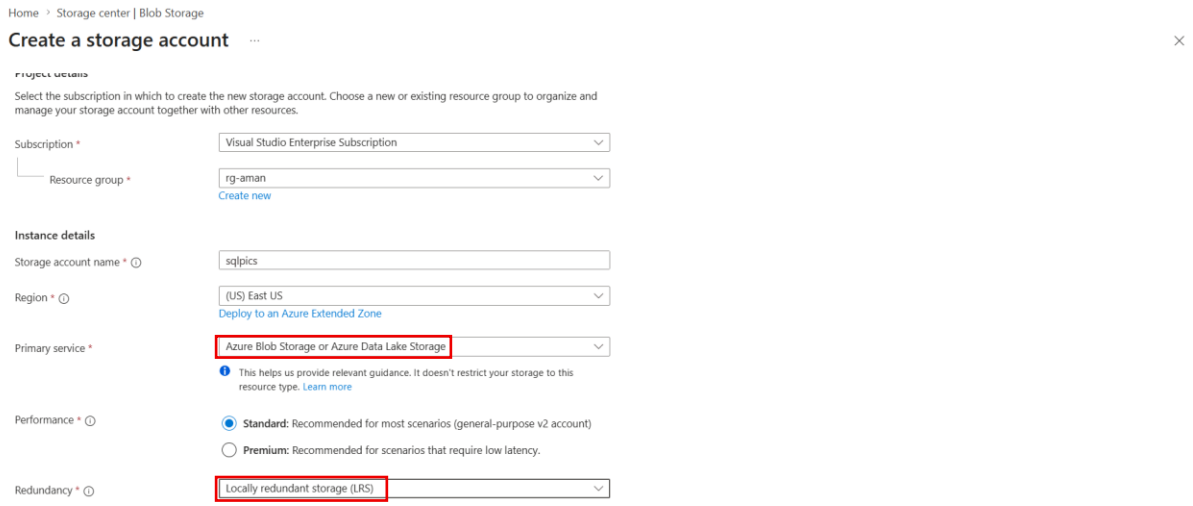
Course ID	Course Image	Course Name	Rating
101		Docker Fundamentals	4.8
102		Docker Compose	4.7
103		Azure for Containers	4.9
104		Kubernetes Basics	4.6

Storing Images in Azure Blob Storage

- Navigate to **Storage Account**
- Create a new **Azure Storage Account** using the same subscription and resource group as the Linux Virtual Machine.

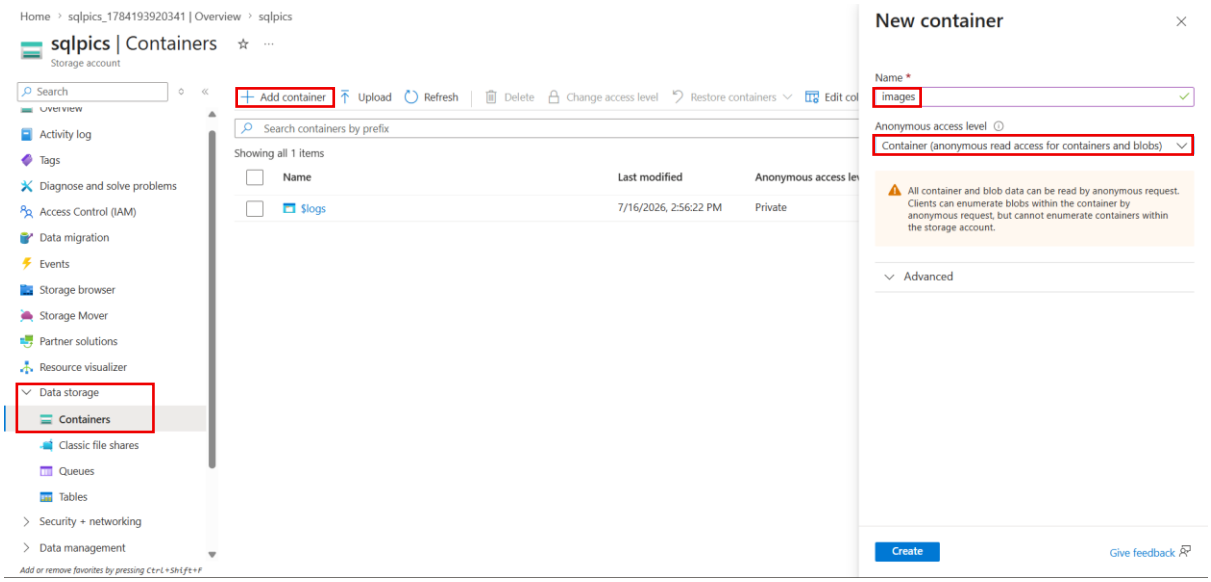


- Provide a unique name for the storage account, select the desired region, choose **Locally Redundant Storage (LRS)**, and complete the deployment.

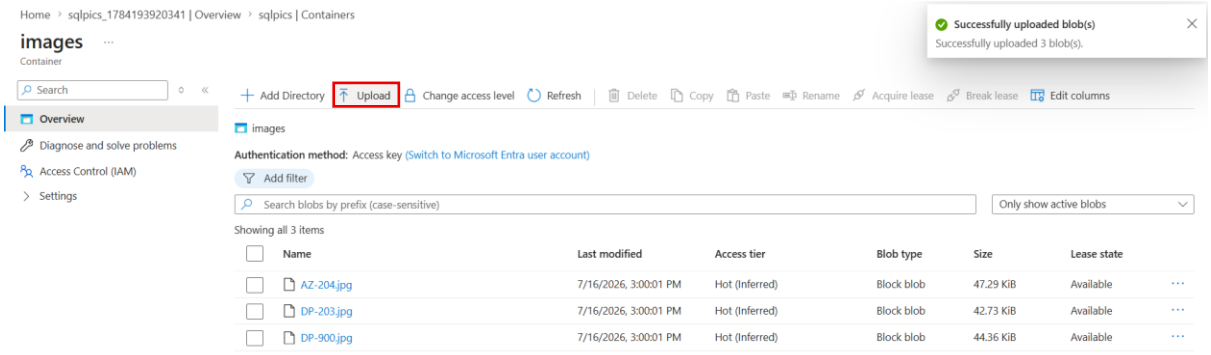


- Open the newly created Storage Account after the deployment is completed.

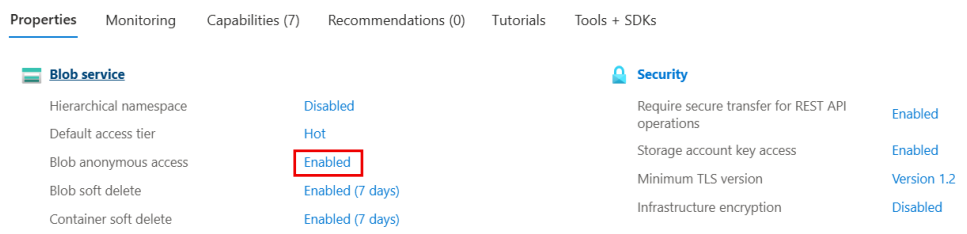
- Navigate to **Containers** under the Data storage and create a new container named **images**.



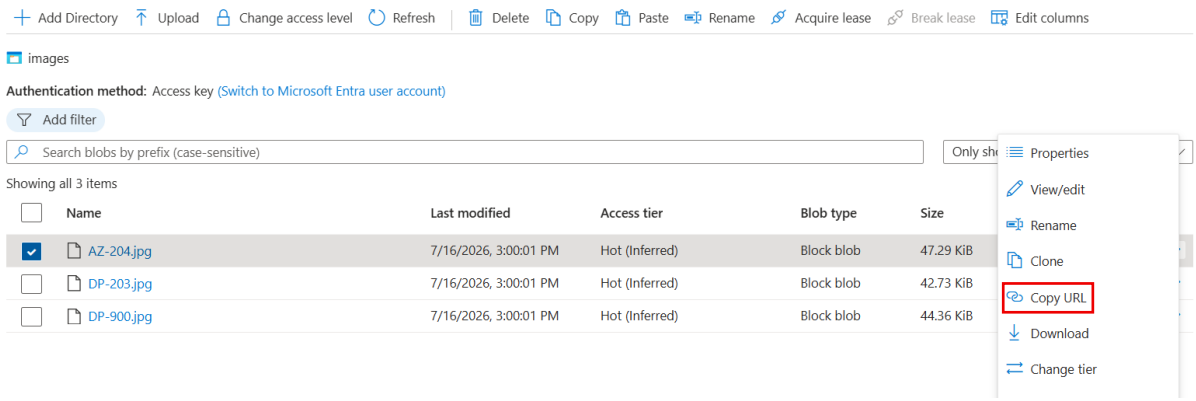
- Open the **images** container and upload the required course image files from the local machine.
- Verify that all image files have been uploaded successfully to the Blob container.



- Go to the **Configuration** section of the Storage Account and enable **Allow Blob Anonymous Access**.
- Save the configuration changes and refresh the Storage Account page.



- Open the **images** container, select **Change Access Level**, and set the access level to **Blob (Anonymous read access for blobs only)**.
- Save the updated access level for the container.
- Open any uploaded image and copy its generated Blob URL.



Deploying the Updated Application with Azure Blob Storage Images

- Connect to the MySQL database and verify that the **Course** table contains the Course ID, Exam Image, Course Name, and Rating columns.

```
sudo docker exec -it mysql-instance mysql -uroot -psqlset1010
```

- Copy the Azure Blob Storage URLs for the uploaded course images from the Azure Storage Account.
- Execute SQL UPDATE statements to replace the existing image paths with the Azure Blob Storage URLs stored in the **ExamImage** column.

```
UPDATE Course
SET ExamImage='https://sqlpics.blob.core.windows.net/images/AZ-204.jpg'
WHERE CourseID=101;

UPDATE Course
SET ExamImage='https://sqlpics.blob.core.windows.net/images/DP-203.jpg'
WHERE CourseID=102;

UPDATE Course
SET ExamImage='https://sqlpics.blob.core.windows.net/images/DP-900.jpg'
WHERE CourseID=103;

UPDATE Course
SET ExamImage='https://sqlpics.blob.core.windows.net/images/AZ-204.jpg'
```

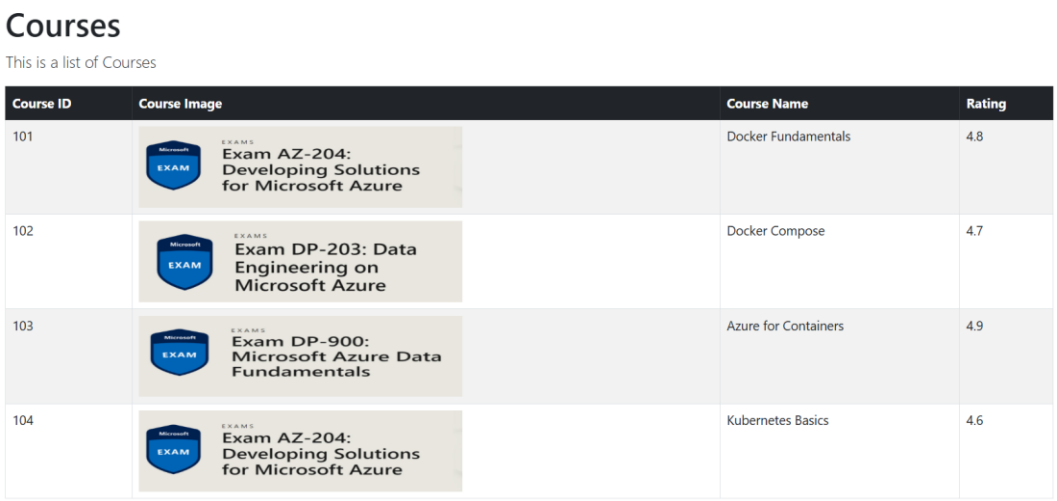
```
WHERE CourseID=104;
```

- Verify that the **ExamImage** column now contains the complete Blob Storage URLs for all course images.

```
SELECT * FROM Course;
```

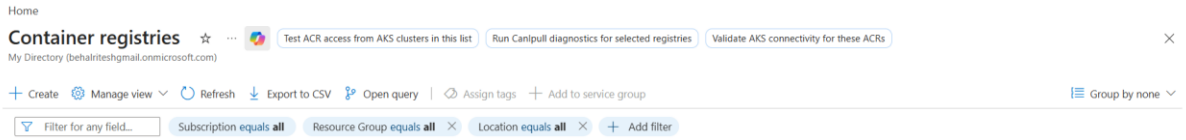
```
mysql> SELECT * FROM Course;
+----------+--------------------------------------+----------------------+--------+
| CourseID | ExamImage                            | CourseName           | rating |
+----------+--------------------------------------+----------------------+--------+
|      101 | https://via.placeholder.com/400x100 | Docker Fundamentals  |    4.8 |
|      102 | https://via.placeholder.com/400x100 | Docker Compose       |    4.7 |
|      103 | https://via.placeholder.com/400x100 | Azure for Containers |    4.9 |
|      104 | https://via.placeholder.com/400x100 | Kubernetes Basics    |    4.6 |
+----------+--------------------------------------+----------------------+--------+
4 rows in set (0.000 sec)

mysql>
```

- Open the PHP application using the Azure Linux VM Public IP address.
- http://<Public-IP>/index.php
- Refresh the application and verify that all course images are successfully loaded from Azure Blob Storage.
- Confirm that the PHP application retrieves the course details from MySQL while displaying the images directly from Azure Blob Storage.

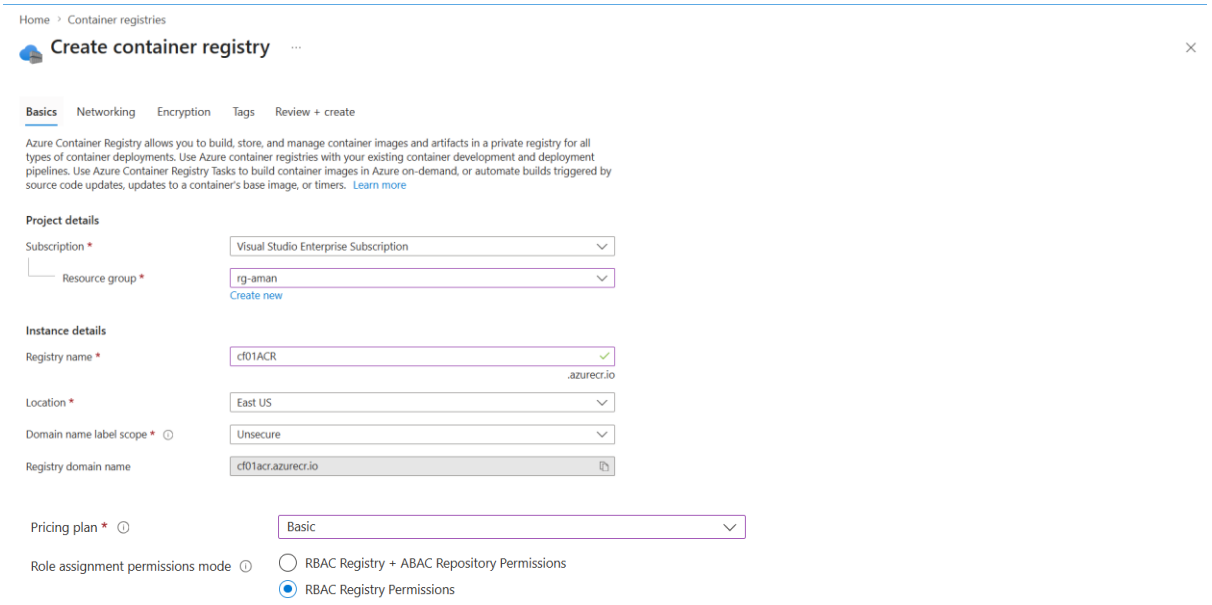


Pushing Docker Images to Azure Container Registry (ACR)

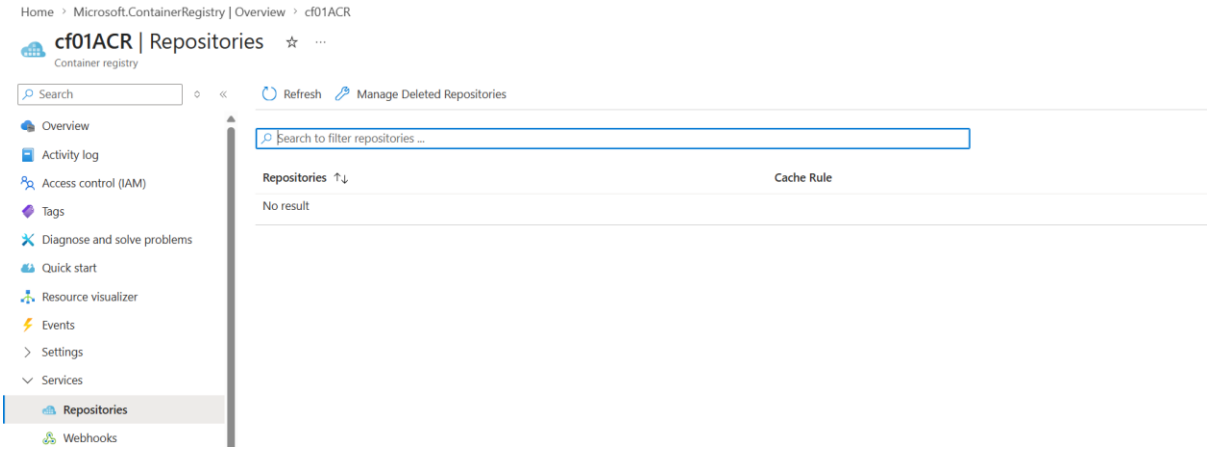
- Create a new **Azure Container Registry (ACR)** resource in the existing Resource Group and choose the desired Azure region.



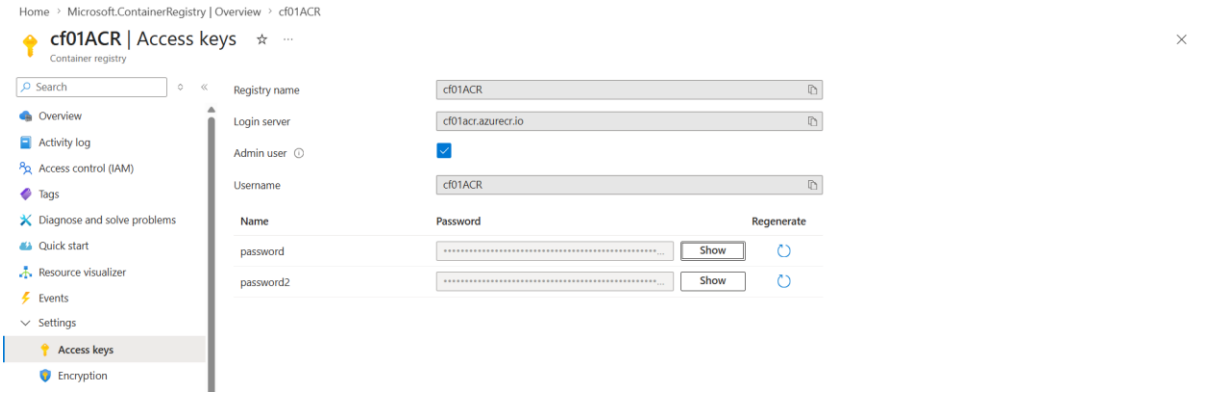
- Provide a unique registry name, keep the default pricing tier (**Basic**), and complete the deployment.



- Open the newly created Azure Container Registry from the Azure Portal.
- Navigate to **Repositories** and verify that no repositories are present initially.



- Go to **Access Keys**, enable the **Admin User**, and note the **Login Server**, **Username**, and **Password**.



- Connect to the Azure Linux Virtual Machine and stop the running Docker Compose application.

```
sudo docker compose down
```

- Log out of Docker Hub.

```
docker logout
```

- Log in to the Azure Container Registry using the Login Server, Username, and Password.

```
docker login <login-server>
```

```
azureuser@cf-linux-vm:~$ sudo docker images

IMAGE          ID             DISK USAGE   CONTENT SIZE   EXTRA
mysql:latest   ae269281abff       1.28GB          288MB
phpapp:v3      fc552b268ff0        725MB          178MB
azureuser@cf-linux-vm:~$ docker logout
Removing login credentials for https://index.docker.io/v1/
azureuser@cf-linux-vm:~$ sudo docker login cf01acr.azurecr.io
Username: cf01ACR
Password:

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
azureuser@cf-linux-vm:~$
```

- Tag the local PHP application image with the Azure Container Registry login server.

```
docker tag phpapp:v3 <login-server>/phpapp:v3
```

- Push the MySQL image to Azure Container Registry.

```
docker push <login-server>/ mysql:latest
```

- Then tag the images:

```
sudo docker tag phpapp:v3 cf01acr.azurecr.io/phpapp:v3

sudo docker tag mysql:latest cf01acr.azurecr.io/mysql:latest
```

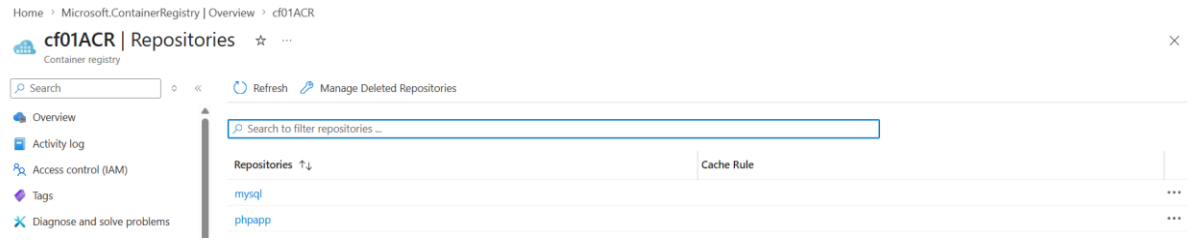
- Push them:

```
sudo docker push cf01acr.azurecr.io/phpapp:v3

sudo docker push cf01acr.azurecr.io/mysql:latest
```

- This is the easiest method because the images are already on the VM.

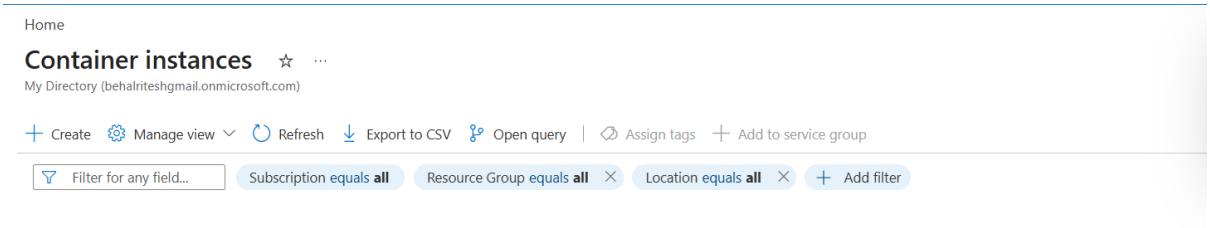
```
azureuser@cf-linux-vm:~$ sudo docker tag phpapp:v3 cf01acr.azurecr.io/phpapp:v3
azureuser@cf-linux-vm:~$ sudo docker tag mysql:latest cf01acr.azurecr.io/mysql:latest
azureuser@cf-linux-vm:~$ sudo docker push cf01acr.azurecr.io/phpapp:v3
The push refers to repository [cf01acr.azurecr.io/phpapp]
e7daf0503344: Pushed
092a953801a3: Pushed
e5587406c34b: Pushed
b00eb207193a: Pushed
d3b60191b176: Pushed
062e450697fa: Pushed
3893442b8746: Pushed
180d13d99fe1: Pushed
8b3caeae61af: Pushed
8b192518ec74: Pushed
91e54e4d2883: Pushed
17ff64b82f58: Pushed
d4d82b592474: Pushed
b1ff61be1778: Pushed
5d1bd08eeb06: Pushed
4f4fb700ef54: Pushed
1468762af3b3: Pushed
f916e83bd43c: Pushed
2e0c3120a4ef: Pushed
v3: digest: sha256:fc552b268ff08bb6efc5ad5d26e6569e3c692c8895156721f555bbb6a4776de0 size: 856
azureuser@cf-linux-vm:~$ sudo docker push cf01acr.azurecr.io/mysql:latest
The push refers to repository [cf01acr.azurecr.io/mysql]
04eb080c2a10: Pushed
ecf47111addb: Pushed
d4c51daaa784: Pushed
9f5b662d2d4d: Pushed
890dd41ed99a: Pushed
aeff22d41055: Pushed
446597dc68d2: Pushed
ad18077cf381: Pushed
1257e53bba2a: Pushed
6b6fad81c717: Pushed
latest: digest: sha256:91e42ae952f37b724f945c734cf067c65aa66bf15d4992a7f7c174918d15fc77 size: 2856
```

- Refresh the **Repositories** page in Azure Container Registry and verify that both the **phpapp** and **mysql** repositories are available.

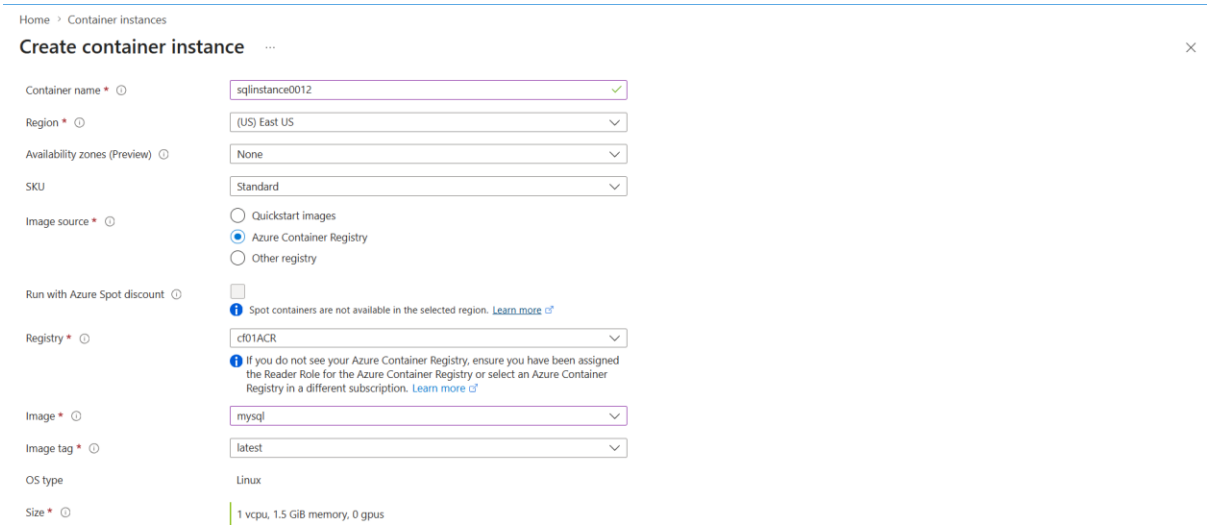


Deploying a Container using Azure Container Instances (ACI)

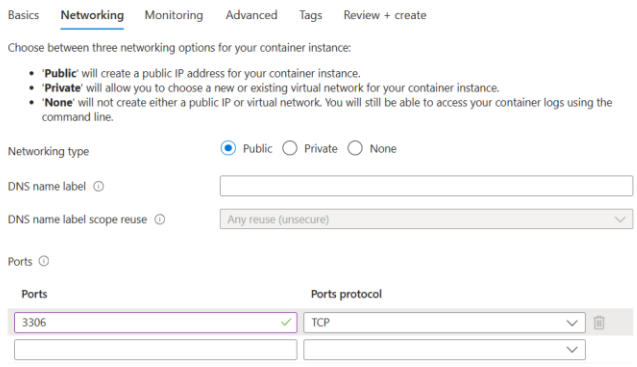
- In the Azure Portal, create a new **Azure Container Instance** resource.



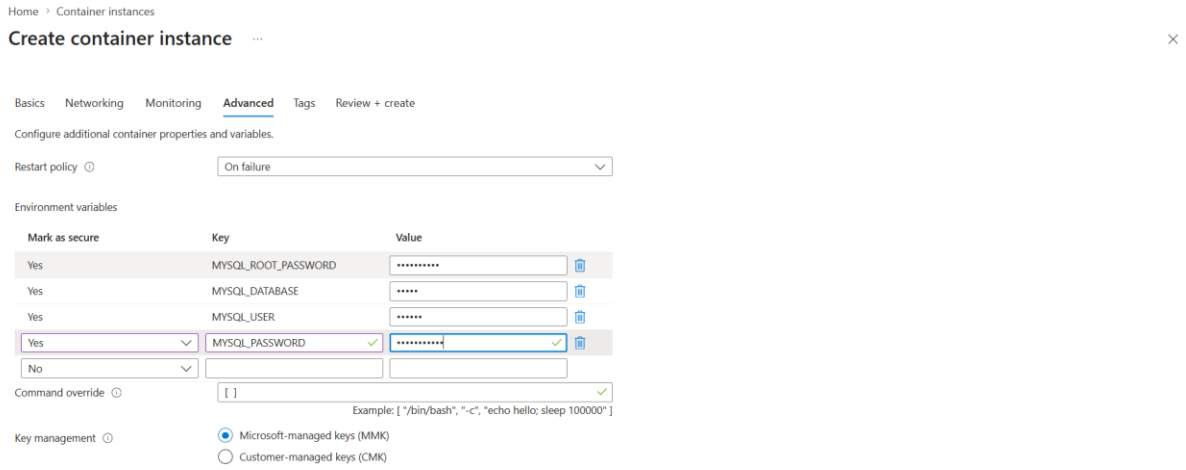
- Select the existing **Resource Group**, provide a unique **Container Name**, and choose the desired Azure region.
- Under **Image Source**, select **Azure Container Registry (ACR)** and choose the previously pushed **MySQL** image (mysql:latest in your case).
- Configure the container with **1 vCPU** and **1.5 GB Memory**, leaving the remaining settings at their default values.



- In the **Networking** tab, select **Public** for network access and expose **Port 3306** to allow MySQL connections.

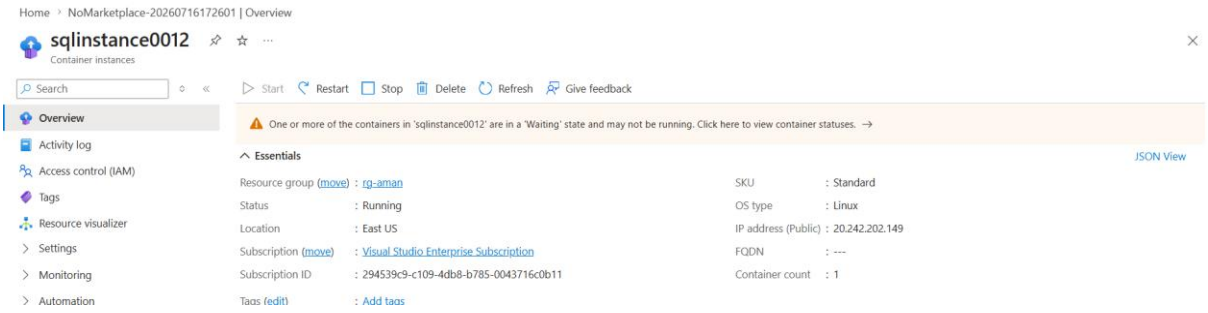


- In advance configure the environments



- Complete the remaining configuration using the default settings and click **Review + Create**, followed by **Create** to deploy the container instance.

- After deployment, open the **Container Instance** resource and verify that the container is in the **Running** state.



- Open the **Containers** section to view the container status, properties, events, and logs generated by the running container.
- Copy the **Public IP Address** assigned to the Azure Container Instance.
- Open **MySQL Workbench**, edit an existing MySQL connection (or create a new one), replace the hostname with the **Public IP Address** of the container instance, and connect using the configured MySQL credentials.